Linear Models II

Xiao-Hua Zhou, Ph.D

PKU Endowed Chair Professor, Beijing International Center for Mathematical Research, and Chair, Department of Biostatistics, Peking University

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Multiple Linear Regression

John Rice

- We have n observations, for each of which we have p predictor measurements and a response measurement.
- Want to develop a model of the form:

$$y_i = \beta_0 + \beta_1 X_{i1} + \ldots + \beta_p X_{ip} + \epsilon_i.$$

  Here ϵ_i is a noise term associated with the ith observation.
- Must estimate $\beta_0, \ldots, \beta_p$.
- Matrix notation:

$$\mathbf{Y} = \mathbf{X}\beta + \epsilon,$$

  where $\mathbf{Y} = (Y_1, \ldots, Y_n)'$, and $\beta = (\beta_0, \beta_1, \ldots, \beta_p)'$.

What makes a Model Linear

- A linear model is linear in the regression coefficients.
- This is a linear model:

$$Y_i = \beta_0 + \beta_1 sin(X_{i1}) + \beta_2 X_{i2} X_{i3} + \epsilon_i.$$

- This is not a linear model:

$$Y_i = \beta_0 + sin(\beta_1 X_{i1}) + \beta_2 X_{i2} + \beta_3 X_{i3} + \epsilon_i.$$

Some Important Questions to Answer

- Is at least one of the predictors $X_1, \ldots, X_p$ useful in predicting the response?
- Do all the predictors help to explain Y, or is only a subset of the predictors useful?
- How well does the model fit the data?
- Given a set of predictor values, what response value should we predict, and how accurate is our prediction?

Least Squared Estimator

- $$\hat{\beta} = argmin_{\beta} RSS(\beta) = argmin_{\beta} (\mathbf{y} - \mathbf{X}\beta)^T (\mathbf{y} - \mathbf{X}\beta),$$

  where $\mathbf{y}$ is the observed value of $\mathbf{Y}$ in the sample.
- By assuming $\mathbf{X}$ is full rank, we can show that

$$\frac{\partial RSS(\beta)}{\partial \beta} = -2\mathbf{X}^T(- \mathbf{X}\beta)$$

$$\frac{\partial RSS(\theta)}{\partial \beta \beta^T} = 2\mathbf{X}^T\mathbf{X}.$$

- $$\widehat{\beta} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}.$$

Least Squared Estimator

- Theorem C. Let X be a random n vector with mean μ and Σ, and let A be a symmetric $n \times n$ matrix. Then,

$$E(X'AX) = trace(A\Sigma) + \mu' A\mu.$$

-

$$Cov(\widehat{\beta}) = \sigma^2 (X'X)^{-1}$$

- Vector of residuals,

$$\widehat{e} = (I - H)Y,$$

where $H = X(X'X)^{-1}X'$. Note that H and $I - H$ are idempotent.

Least Squared Estimator

- Sum of squared residuals:

$$\sum_{i=1}^{n}(Y_i - \widehat{Y}_i)^2 = || \ Y - \widehat{Y} \ ||^2 = || \ (I - H)Y \ ||^2 = \ Y'(I - P)Y$$

- $$E(Y'(I - P)Y) = (E(Y))'(I_n - P)E(Y) + \sigma^2 trace(I - P)$$

- $$trace(I - P) = trace(I_n) - trace(P) =$$
$$n - trace(X'X(X'X)^{-1}) = n - p,$$

  because $trace(AB) = trace(BA)$

Least Squared Estimator

- Vector of residuals, $\widehat{e} = Y - \widehat{Y}$, has the covariance matrix, $\sigma^2(I - H)$.
- If the errors have the covariance matrix $\sigma^2 I$, the residuals are uncorrelated with the fitted values, because $Cov(\widehat{e}, \widehat{Y}) = Cov((I - H)Y, HY) = (I - H)Cov(Y, Y)H' = \sigma^2(I - H)H' = 0$.

Least Squared Estimator

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Prediction with LS regression

- We first the linear model with a training set of observations.
- We get coefficient estimates $\widehat{\beta}_1, \ldots, \widehat{\beta}_p$.
- Then, we obtain predictions using our model for a new subject with $x_0 = (x_{01}, \ldots, x_{0p})$, of the form

$$\widehat{y}_0 = \widehat{\beta}_1 x_{01} + \ldots + \widehat{\beta}_p x_{0p}.$$

- We can evaluate the training error, i.e. the extent to which the model fits the observations used to train it.

Prediction with LS regression

- One way to quantify the training error is using the mean squared error (MSE):

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \widehat{y}_i)^2 = \frac{1}{n}\sum_{i=1}^{n}(y_i - (\widehat{\beta}_1 X_{i1} + \ldots + \widehat{\beta}_p X_{ip}))^2$$

  where $\widehat{y}_i = \widehat{\beta}_1 X_{i1} + \ldots + \widehat{\beta}_p X_{ip}$.

- Training error is closely related to the R^2 for a linear model; that is, the proportion of variance explained.
- Big $R^2 \Leftrightarrow$ Small Training Error.

Geometrical Interpretation of Prediction

- The fitted values at the training inputs are:

$$\widehat{\mathbf{y}} = \mathbf{X}\widehat{\beta} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}.$$

- $\mathbf{H} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T$ is called projection matrix.
- We minimize $RSS(\beta) = || \mathbf{y} - \mathbf{X}\beta ||^2$ by choosing β so that the residual vector $\mathbf{y} - \widehat{\mathbf{y}}$ is orthogonal to this subspace, spanned by $\mathbf{X}$.
- The resulting estimate $\mathbf{y}$ is hence the orthogonal projection of $\mathbf{y}$ onto this subspace.

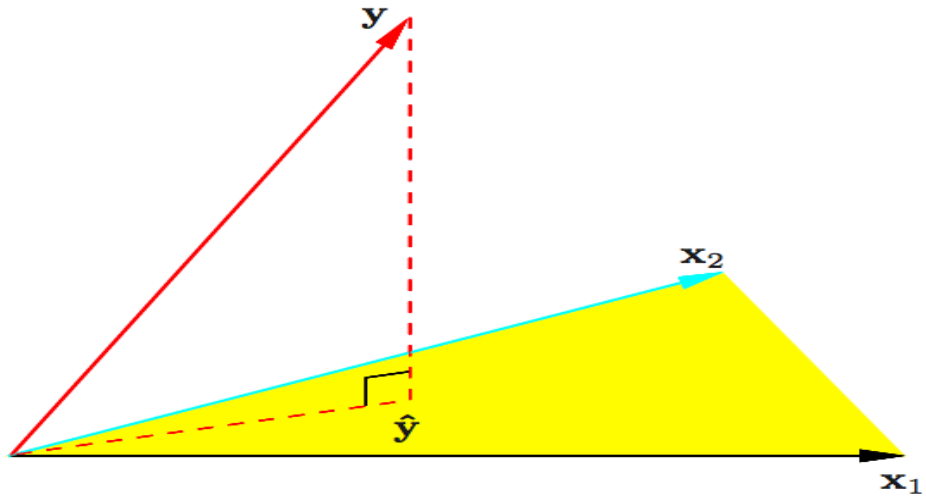


FIGURE 3.2. *The N-dimensional geometry of least squares regression with two*

Geometrical Interpretation of Prediction

- It might happen that the columns of X are not linearly independent, so that X is not of full rank.
- This would occur, for example, if two of the inputs were perfectly correlated, (e.g., $x_2 = 3x_1$). Then $\mathbf{X}^T\mathbf{X}$ is singular and the least squares coefficients $\widehat{\beta}$ are not uniquely defined.

Geometrical Interpretation of Prediction

- However, the fitted values $\mathbf{y} = \mathbf{X}\widehat{\beta}$ are still the projection of $\mathbf{y}$ onto the column space of $\mathbf{X}$; there is just more than one way to express that projection in terms of the column vectors of $\mathbf{X}$.
- The non-full-rank case occurs most often when one or more qualitative inputs are coded in a redundant fashion.
- There is usually a natural way to resolve the non-unique representation, by recoding and/or dropping redundant columns in $\mathbf{X}$.

Geometrical Interpretation of Prediction

- Rank deficiencies can also occur in signal and image analysis, where the number of inputs p can exceed the number of training cases N.
- In this case, the features are typically reduced by filtering or else the fitting is controlled by regularization.

Address Some of These Questions

- One: Is There a Relationship Between the Response and Predictors? In the multiple regression setting with p predictors, we need to ask whether all of the regression coefficients are zero
- We test the null hypothesis, $H_0 : \beta_1 = \ldots = \beta_p = 0$. versus the alternative H_1 : at least one β_j is non-zero.
- This hypothesis test is performed by computing the F-statistic,

$$F = \frac{(TSS - RSS)/p}{RSS/(n - p - 1)}.$$

Address Some of These Questions

- If the linear model is true under H_0, one can show that

$$E(RSS/(n-p-1)) = \sigma^2, E((TSS - RSS/p) = \sigma^2.$$

- Hence, when there is no relationship between the response and predictors, one would expect the F-statistic to take on a value close to 1.
- On the other hand, if H_1 is true, then $E((TSS - RSS)/p) > \sigma^2$.
- So we expect F to be greater than 1.

Address Some of These Questions

- We test the null hypothesis, $H_0 : \beta_{p-q+1} = \ldots = \beta_p = 0$.
- In this case we fit a second model that uses all the variables except those last q.
- Suppose that the residual sum of squares for that model is RSS_0.
- This hypothesis test is performed by computing the F-statistic,

$$F = \frac{(RSS_0 - RSS)/q}{RSS/(n-p-1)},$$

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Inference on Model Parameters

- Since we are not really interested in the coefficient estimates based on our single sample, the question is: "what can we say about the population-level parameters based on our estimated coefficients"?
  - is there really an association between shoe size and run time?
  - do height, weight and shoe size provide any information about run time?

Example, continued

- We are interested in the following question: Are the differences in the means of the measurements from the various labs significant, or might they be due to chance?
- We will use the F test to answer this question by assuming an explicit statistical model for the variability seen in the figure above.

Example, continued

- According to this model, there is an unknown mean level associated with each laboratory, and the devitions from this mean level of the 10 measurements within a laboratory are independent, normally distributed, random variables.
- With the aid of this model, we will see whether it is plausible that the unknown laboratory means are all equal, so that the variability between labs displayed in the figure is entirely due to chance.

Example, continued

- Instead of fitting a separate simple linear regression model for each predictor, a better approach is to extend the simple linear regression model so that it can directly accommodate multiple predictors.
- We can do this by giving each predictor a separate slope coefficient in a single model. In general, suppose that we have p distinct predictors.
- Then the multiple linear regression model takes the form:

$$Y = \beta_0 + \beta_1 \text{TV} + \beta_2 \text{ radio } + \beta_3 \text{ newspaper } + \epsilon.$$

Example Result

For the Advertising data, multiple regression results

	Coefficient	Std. error	t-statistic	p-value
Intercept	2.939	0.3119	9.42	< 0.0001
TV	0.046	0.0014	32.81	< 0.0001
Radio	0.189	0.0086	21.89	< 0.0001
newspaper	-0.001	0.0059	-0.18	0.8599

For the Advertising data, simpleregression results

	Coefficient	Std. error	t-statistic	p-value
Intercept	12.351	0.621	19.88	< 0.0001
newspaper	0.055	0.017	3.30	< 0.0001

Example Result, continued

- The simple and multiple regression coefficients can be quite different.
- This difference stems from the fact that in the simple regression case, the slope term represents the average effect of a \$1,000 increase in newspaper advertising, ignoring other predictors such as TV and radio.
- However, in the multiple regression setting, the coefficient for newspaper represents the average effect of increasing newspaper spending by \$1,000 while holding TV and radio fixed.
- Why can the multiple regression suggest no relationship between sales and newspaper while the simple linear regression implies the opposite?

Example Result, continued

Consider the correlation matrix for the three predictor variables and response variable, displayed in Tble below:

	TV	radio	newspaper	sales
TV	1.0000	0.0548	0.0567	0.7822
radio		1.0000	0.3541	0.5762
newspaper			1.0000	0.2283
sales				1.0000

Example Result, continued

- Correlation between radio and newspaper is 0.35.
- Spending more on newspaper advertising, more is spent on radio advertising.
- If the multiple regression is correct and newspaper advertising has no direct impact on sales, but radio advertising does increase sales.
- If we spend more on radio, our sales will tend to be higher. Then, we also tend to spend more on newspaper advertising in those same markets.
- A simple linear regression shows that higher values of newspaper tend to be associated with higher values of sales, even though newspaper advertising does not actually affect sales. So newspaper sales are a surrogate for radio advertising; newspaper gets "credit" for the effect of radio on sales.

Adjusted R^2

- The sample R^2 is an estimate of the population R^2:

$$1 - \frac{var(Y \mid X)}{var(Y)}$$

- Since it is a ratio, the plug-in estimate R^2 is biased, although the bias is not large unless the sample size is small or the number of covariates is large.
-

Adjusted R^2

- The adjusted R^2 is an approximately unbiased estimate of the population R^2:

$$1 - (1 - R^2)\frac{n-1}{n-p-1}.$$

- The adjusted R^2 is always less than the unadjusted R^2 because $(n-1)/(n-p-1) \geq 1$, assuming $n-1-p > 0$.
- The adjusted R^2 is always less than or equal to one, but can be negative.

Potential Problems with Linear Models

James, Witten, Hastie, and Tibshirani (2014). Introduction to Statistical Learning

- Non-linearity of the response-predictor relationships.
- Correlation of error terms.
- Non-constant variance of error terms.
- Outliers.
- High-leverage points.
- Collinearity.

In practice, identifying and overcoming these problems is as much an art as a science.

Non-linearity

- The linear regression model assumes that there is a straight-line relationship between the predictors and the response. If the true relationship is far from linear, then our conclusion may be wrong.
- Residual plots are a useful graphical tool for identifying non-linearity.
- Given a simple linear regression model, we can plot the residuals, $e_i = y_i - \widehat{y}_i$, versus the predictor x_i. In the case of a multiple regression model.

Non-linearity

- Given a simple linear regression model, we can plot the residuals, $e_i = y_i - \widehat{y}_i$, versus the predictor x_i. In the case of a multiple regression model.
- For a multiple linear regression, we plot e_i versus the fitted values $\widehat{y}_i$.
- Ideally, the residual plot will show no discernible pattern. If there is a suspected pattern, some aspect of the linear model may not be true.

Example

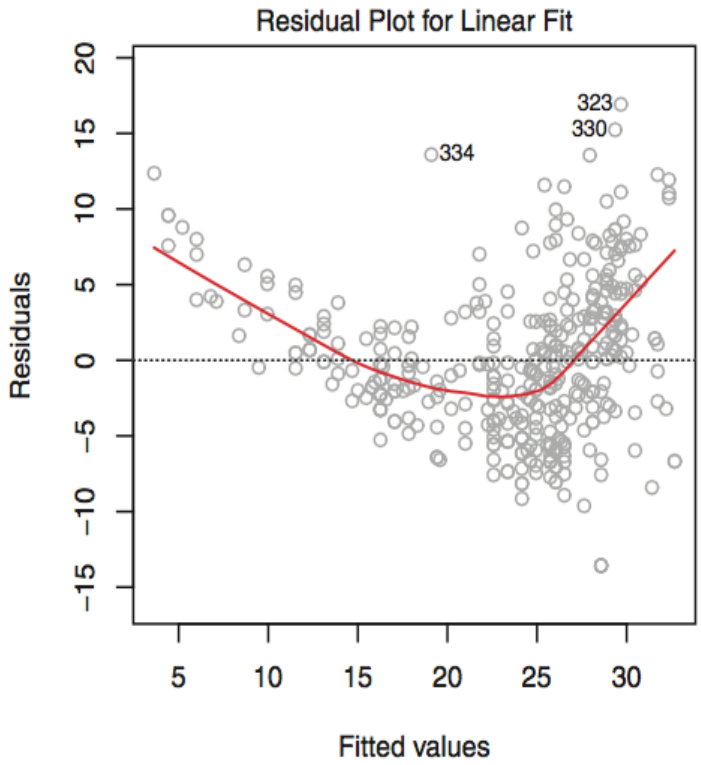
We are interested in studying the relationship between Gas mileage and horsepower of cars. We fit two linear regression models:

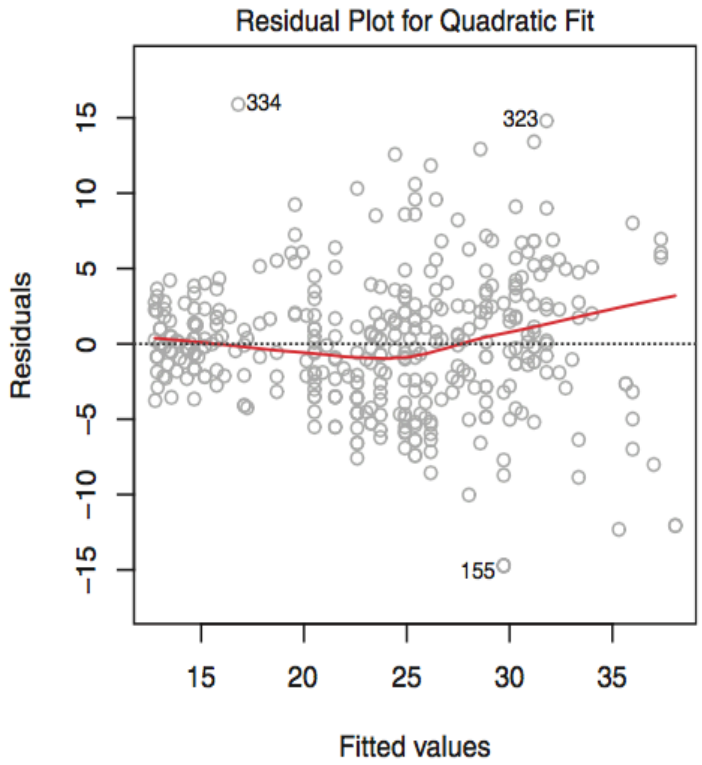
- A linear regression of mpg on horsepower:

$$mpg = \beta_0 + \beta_1 \times horsepower + \epsilon$$

- A linear regression of mpg on horsepower and horsepower^2:

$$mpg = \beta_0 + \beta_1 \times horsepower + \beta_2 horsepower^2 + \epsilon$$


Residual Plot for Linear Fit
Residuals
Fitted values
323
330
334
20
15
10
5
0
−5
−10
−15
5
10
15
20
25
30


Residual Plot for Quadratic Fit
Residuals
Fitted values
334
323
155
15
10
5
0
−5
−10
−15
15
20
25
30
35

Example, continued

- The left panel of Figure above displays a residual plot from the linear regression of mpg onto horsepower on the Auto data set. The red line is a smooth fit to the residuals, which is displayed in order to make it easier to identify any trends. The residuals exhibit a clear U-shape, which provides a strong indication of non-linearity in the data.
- The right-hand panel of Figure displays the residual plot that results from the second model, which contains a quadratic term. There appears to be little pattern in the residuals, suggesting that the quadratic term improves the fit to the data.

Example, continued

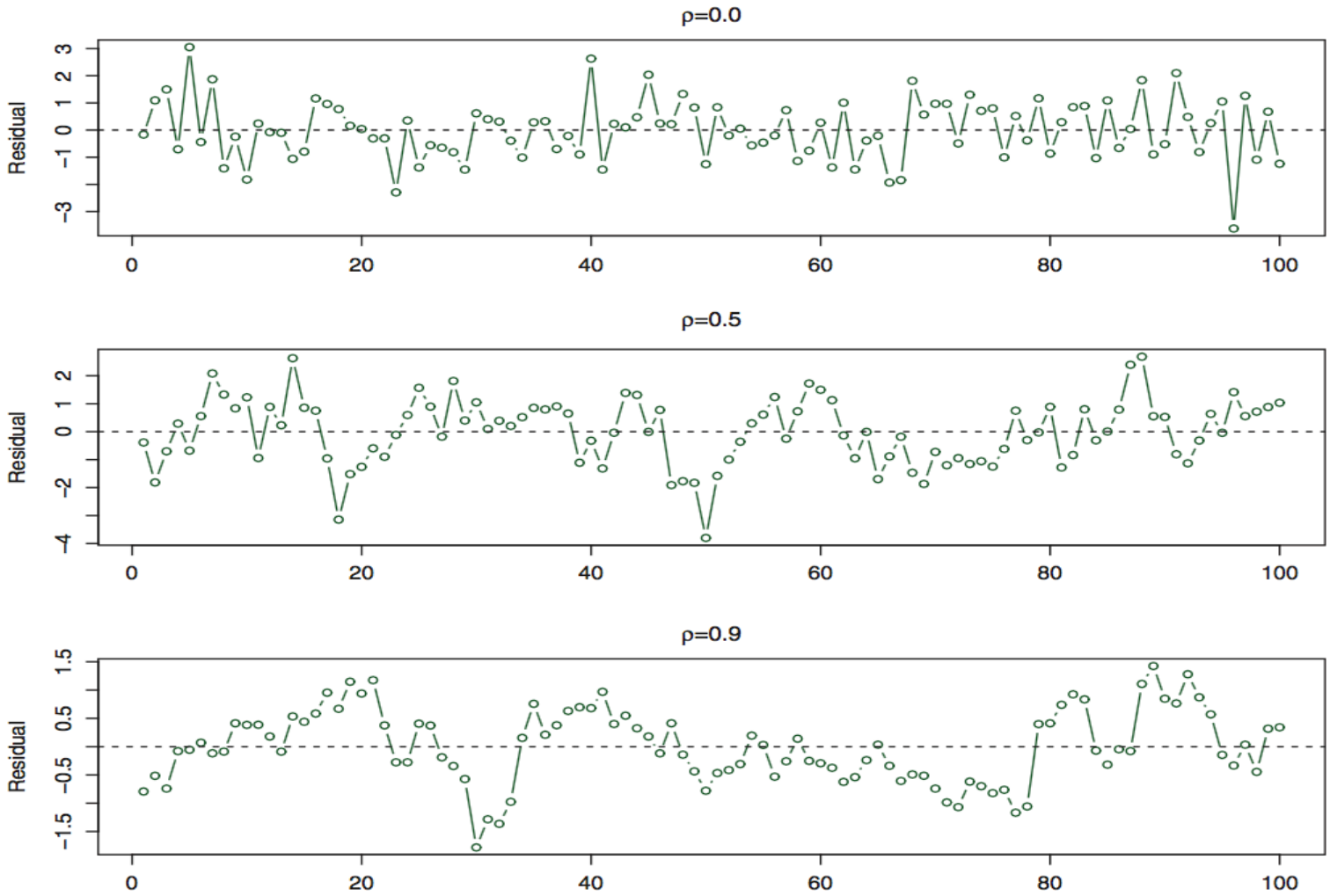
- If the residual plot indicates that there are non-linear associations in the data, then a simple approach is to use non-linear transformations of the predictors, such as $logX$, $\sqrt{X}$, and X^2, in the regression model.

Correlation of Error Terms

- Another important assumption of the linear regression model: the error terms, $\epsilon_1, \ldots, \epsilon_n$, are uncorrelated.
- This assumption means that the fact that ϵ_i is positive provides little or no information about the sign of ϵ_{i+1}

Residual Plots

Plots of residuals from simulated time series data sets generated with differing levels of correlation ρ between error terms for adjacent time points.



Correlation of Error Terms

- Methods are available to properly take account of correlations in the error terms in time series data.
- Correlation among the error terms can also occur outside of time series data. For instance, consider a study in which individuals' blood pressues are predicted from their weights.
- The assumption of uncorrelated errors could be violated if some of the individuals in the study are members of the same family, or eat the same diet, or have been exposed to the same environmental factors.

Correlation of Error Terms

- In general, the assumption of uncorrelated errors is extremely important for linear regression as well as for other statistical methods.

Heteroscedasticity

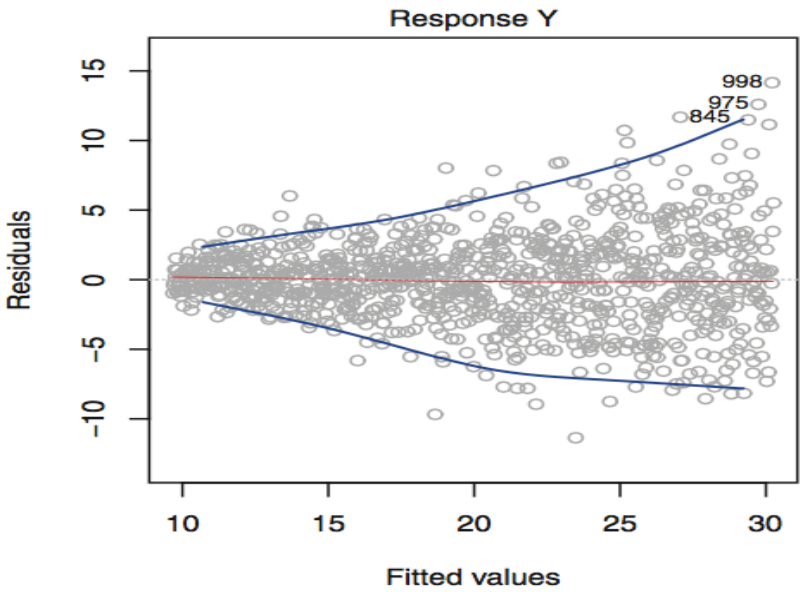
- Another important assumption of the linear regression model is that the error terms have a constant variance, $Var(\epsilon_i) = \sigma^2$.
- The standard errors, confidence intervals, and hypothesis tests associated with the linear model rely upon this assumption.
- However, it is often the case that the variances of the error terms are non-constant. For example, the variances of the error terms may increase with the value of the response.

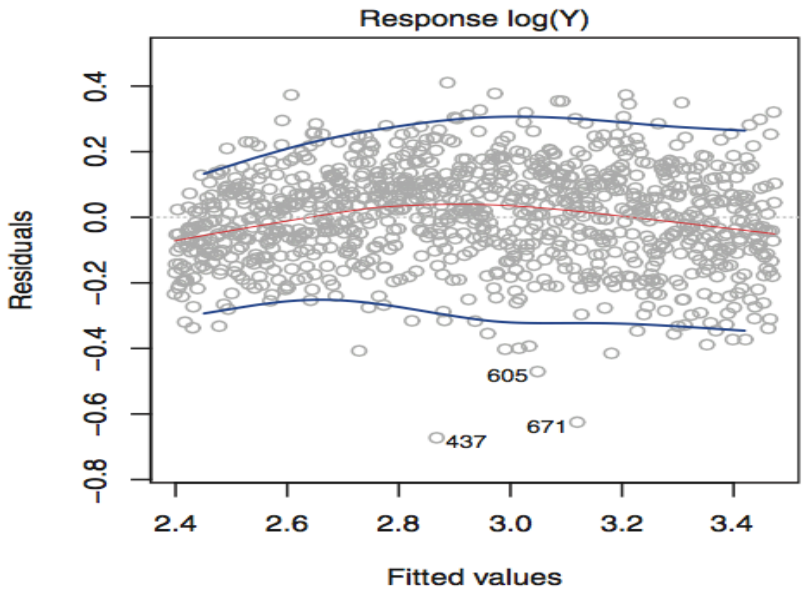
Heteroscedasticity

- One can identify non-constant variances in the errors, or heteroscedasticity, from the presence of a funnel shape in the residual plot.
- See the example below.

Residual Plots

Residual plots. In each plot, the red line is a smooth fit to the residuals, intended to make it easier to identify a trend.





Heteroscedasticity

- When faced with this problem, one possible solution is to transorm the response Y using a concave function such as $logY$ or $\sqrt{Y}$.
- Such a transformation results in a greater amount of shrinkage of the larger responses, leading to a reduction in heteroscedasticity.

Heteroscedasticity

- Sometime, we have a good idea of the variance of each response. For example, if each of these raw observations is uncorrelated with variance 2, then their average has variance $\sigma_i^2 = \sigma^2/n_i$.
- In this case a simple remedy is to fit our model by weighted least squares, with weights proportional to the inverse variances—i.e. $w_i = n_i$.

Outliers

- An outlier is a point for which y_i is far from the value predicted by the model.
- Outliers can arise for a variety of reasons, such as incorrect recording of an observation during data collection.
- Residual plots can be used to identify outliers. In this example, the outlier is clearly visible in the residual plot illustrated in the center panel of Figure below.

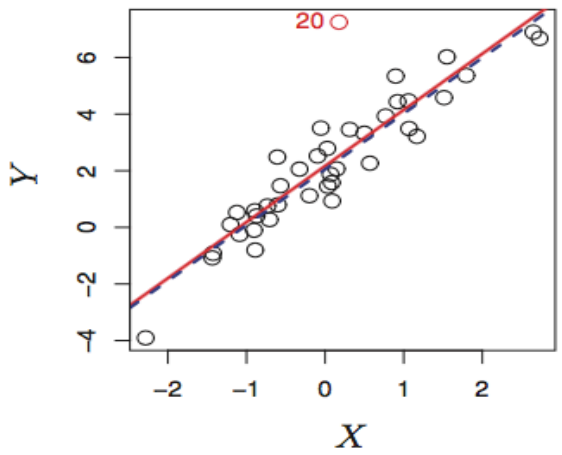
Outliers

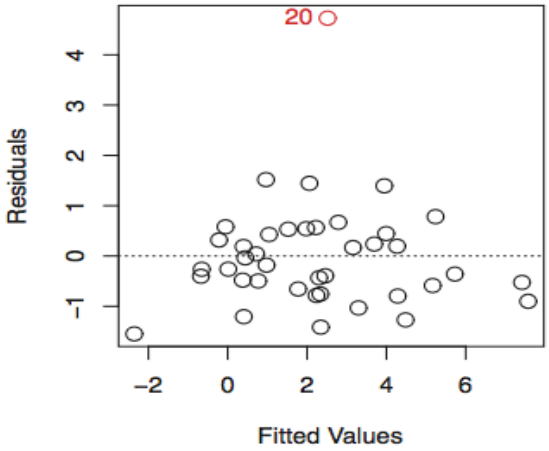
- But in practice, it can be difficult to decide how large a residual needs to be before we consider the point to be an outlier.
- To address this problem, instead of plotting the residuals, we can plot the studentized residuals, computed by dividing each residual e_i by its estimated standard error.

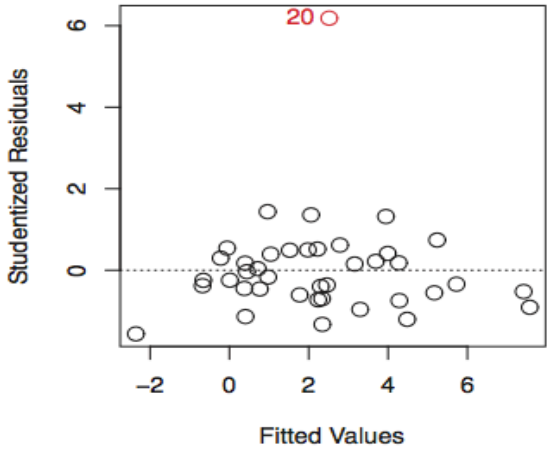
Outliers

- Observations whose studentized residuals are greater than 3 in absolute value are possible outliers.
- If we believe that an outlier has occurred due to an error in data collection or recording, then one solution is to simply remove the observation.
- However, care should be taken, since an outlier may instead indicate a deficiency with the model, such as a missing predictor.

Residual Plots







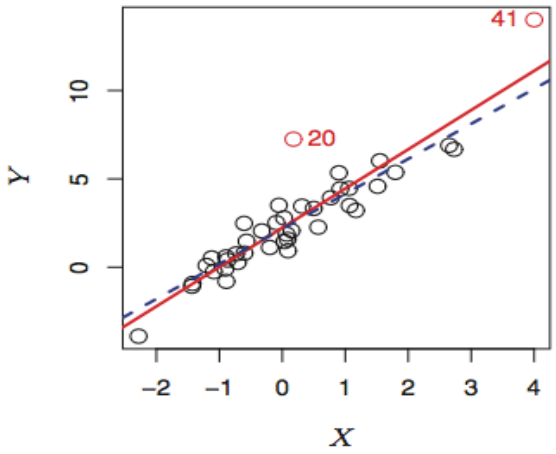
High Leverage Points

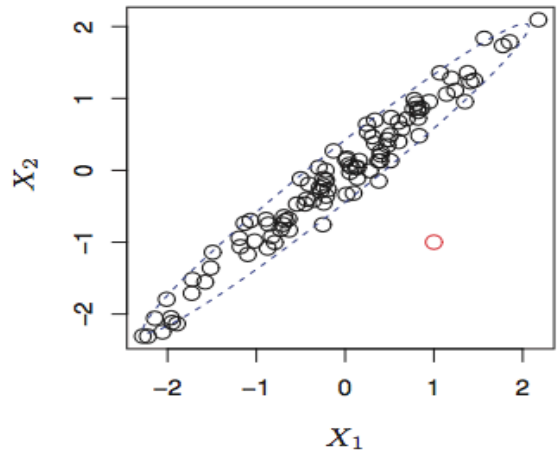
- We just saw that outliers are observations for which the response y_i is unusual given the predictor x_i.
- In contrast, observations with high leverage high leverage have an unusual value for x_i.
- For example, observation 41 in the left-hand panel of Figure below has high leverage, in that the predictor value for this observation is large relative to the other observations.

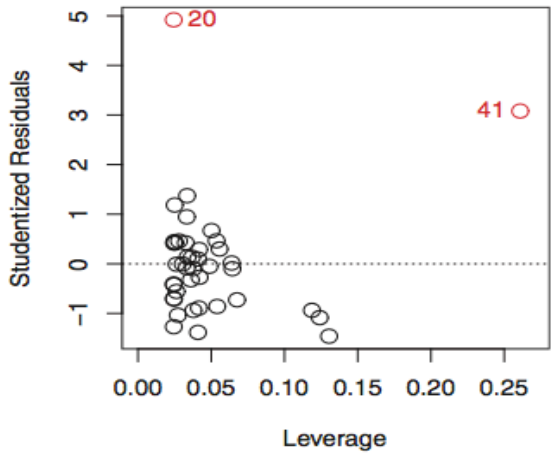
High Leverage Points

- The red solid line is the least squares fit to the data, while the blue dashed line is the fit produced when observation 41 is removed.
- Comparing the left-hand panels of Figures below , we observe that removing the high leverage observation has a much more substantial impact on the least squares line

Residual Plots







High Leverage Points

- In fact, high leverage observations tend to have a sizable impact on the estimated regression line.
- It is cause for concern if the least squares line is heavily affected by just a couple of observations, because any problems with these points may invalidate the entire fit.
- For this reason, it is important to identify high leverage observations.
- In a simple linear regression, high leverage observations are fairly easy to identify, since we can simply look for observations for which the predictor value is outside of the normal range of the observations.

High Leverage Points

- But in a multiple linear regression with many predictors, it is possible to have an observation that is well within the range of each individual predictor's values, but that is unusual in terms of the full set of predictors.
- An example is shown in the center panel of Figure above, for a data set with two predictors, X_1 and X_2.

Extension of Leverage Statistic to Multiple Covariates

- Recall the LSE of β can be written as follows:

$$\widehat{y} = Hy,$$

  where $H = X(X'X)^{-1}X'$, which is $n \times n$ projection matrix.
- Let $H = (h_{ij})$. If h_{ij} is large, the ith observation has a substantial impact on the jth fitted value.

Extension of Leverage Statistic to Multiple Covariates

- Since H is symmetric and idempotent, the diagonal entries represent both the ith row and ith column ($H^T H = H$, also $I - H$ is idempotent):

$$h_i = h_i' h_i = \sum_{j=1}^{n} h_{ij}^2,$$

  which implies that $h_i = h_{ii}$.
- In order to quantify an observation's leverage, we compute the leverage statistic, h_i.
- A large value of this statistic indicates an observation with high leverage leverage.

Leverage Statistic

- The average hat-value is: $\bar{h} = (k+1)/n$.
- Hat values are bounded between $1/n$ and 1.
- For a simple linear regression, statistic

$$h_i = \frac{1}{n} + \frac{(x_i - \bar{x})^2}{\sum_{j=1}^{n}(x_j - \bar{x})^2},$$

  which measures distance from the mean of X.
- In multiple regression, h_i measures the distance from the centroid point of $X's$ (point of means).

Rule of Thumb

- Hat values exceeding about twice the average hat-value should be considered noteworthy
- With large sample sizes, however, this cut-off is unlikely to identify any observations regardless of whether they are true leverage points.

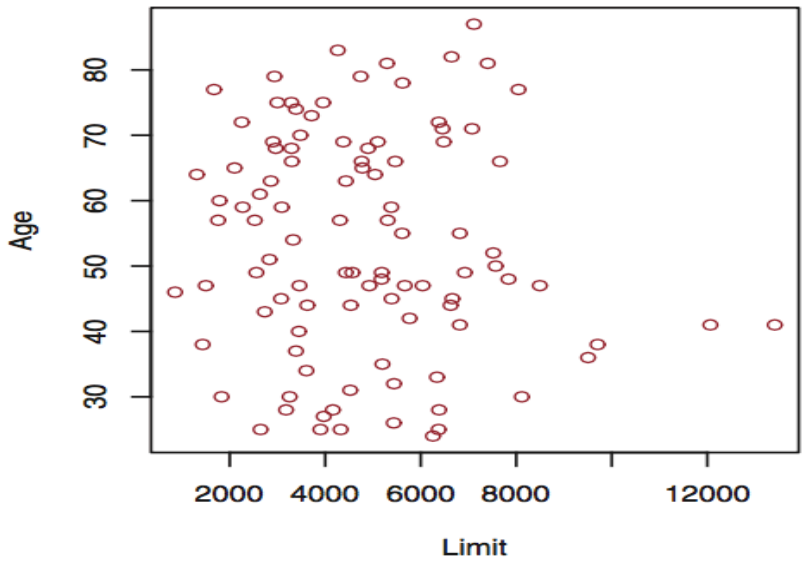
Leverage Statistic

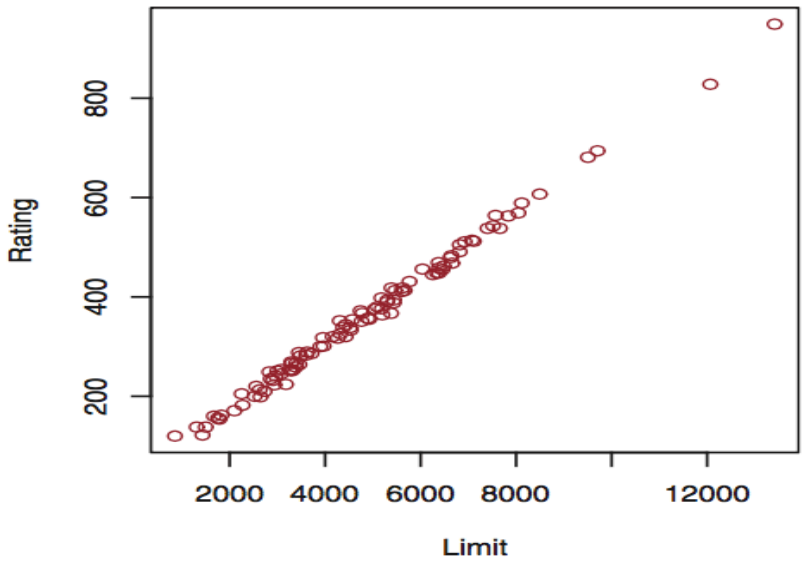
- The right-hand panel of Figure above provides a plot of the studentized residuals versus hi for the data in the left-hand panel of Figure.
- Observation 41 stands out as having a very high leverage statistic as well as a high studentized residual. In other words, it is an outlier as well as a high leverage observation. This is a particularly dangerous combination.
- This plot also reveals the reason that observation 20 had relatively little effect on the least squares fit in Figure: it has low leverage.

Collinearity

- Collinearity refers to the situation in which two or more predictor variables are closely related to one another.

Collinearity





Collinearity

- A simple way to detect collinearity is to look at the correlation matrix of the predictors.
- An element of this matrix that is large in absolute value indicates a pair of highly correlated variables, and therefore a collinearity problem in the data.

Collinearity

- However, not all collinearity problems can be detected by inspection of the correlation matrix: it is possible for collinearty to exist between three or more variables even if no pair of variables has a particularly high correlation.
- We call this situation multicollinearity.
- Instead of inspecting the correlation matrix, a better way to assess multi- collinearity is to compute the variance inflation factor (VIF).

Collinearity

- The VIF is the ratio of the variance of $\widehat{\beta}_k$ when fitting the full model divided by the variance of $\widehat{\beta}_k$ if fit on its own.
- The smallest possible value for VIF is 1, which indicates the complete absence of collinearity.
- Typically in practice there is a small amount of collinearity among the predictors.
- As a rule of thumb, a VIF value that exceeds 5 or 10 indicates a problematic amount of collinearity.

Collinearity

- A variance inflation factor (VIF) quantifies how much the variance of the estimated coefficient is inflated.
- So, the variance inflation factor for the estimated coefficient $\widehat{\beta}_k$, denoted by VIF_k - is just the factor by which the variance is inflated.

Collinearity

- Assume that we have a model with only x_k as the predictor:

$$Y_i = \beta_0 + \beta_k x_{ik} + \epsilon_i$$

Then, we have shown that the variance of the estimated coefficient $\widehat{\beta}_k$:

$$Var(\widehat{\beta}_k)_{min} = \frac{\sigma^2}{\sum_{i=1}^{n}(x_{ik} - \bar{x}_k)^2}.$$

- It can be shown that (Homework)

$$Var(\widehat{\beta}_k) = \frac{\sigma^2}{\sum_{i=1}^{n}(x_{ik} - \bar{x}_k)^2} \times \frac{1}{1 - R_k^2}$$

where R_k^2 is the R^2-value obtained by repressing the *kth* predictor on the remaining predictors.

Collinearity

- Therefore,
$$\frac{Var(\widehat{\beta}_k)}{Var(\widehat{\beta}_k)_{min}} = \frac{1}{1 - R_k^2}.$$
- Hence,
$$VIF_k = \frac{1}{1 - R_k^2}.$$
- If R_k^2 is close to one, then collinearity is present, and so the VIF will be large.

Example on Customer default records for a credit card company

Variables:

- Outcome: balance is average credit card debt for an individuals).
- Predictors: age, cards (number of credit cards), education (years of education), income (in thousands of dollars), limit (credit limit), and rating (credit rating).

Example on Customer default records for a credit card company

ing

		Coefficient	Std. error	t-statistic	p-value
Mod1	Intercept	-173.411	43.828	-3.957	< 0.0001
	age	2.292	0.672	-3.407	0.0007
	limit	0.173	0.005	34.496	< 0.0001
Mod2	Intercept	-377.537	45.254	-8.343	< 0.0001
	rating	2.202	0.952	2.312	0.0213
	limit	0.025	0.064	0.384	0.702

Example on Customer default records for a credit card company

- Table above compare coefficient estimates obtained from two separate multiple regression models.
- The first is a regression of balance on age and limit, and the second is a regression of balance on rating and limit.
- In the first regression, both age and limit are highly significant with very small p-values.

Example on Customer default records for a credit card company

- In the second, the collinearity between limit and rating has caused the standard error for the limit coefficient estimate to increase by a factor of 12 and the p-value to increase to 0.701.
- In other words, the importance of the limit variable has been masked due to the presence of collinearity.
- To avoid such a situation, it is desirable to identify and address potential collinearity problems while fitting the model.

Solution to Collinearity

- When faced with the problem of collinearity, there are two simple solutions. The first is to drop one of the problematic variables from the regression.
- The second solution is to combine the collinear variables together into a single predictor.

Introduction to R

The best way to learn a new language is to try out the commands. R can be downloaded from

`http://cran.r-project.org/`

R uses functions to perform operations. To run a function called *funcname*, we type *funcname*($input_1$, $input_2$), where the inputs (or arguments) $input_1$ and $input_2$ tell R how to run the function. A function can have any number of inputs. For example, to create a vector of numbers, we use the function *c*() (for concatenate).

The following command instructs R to join together the numbers $1, 3, 2$, and 5, and to save them as a vector named x. When we type x, it vector gives us back the vector.

```
> x <- c(1 ,3 ,2 ,5)
```

Introduction to R

- Note that the $>$ is not part of the command; rather, it is printed by R to indicate that it is ready for another command to be entered. We can also save things using $=$ rather than $<-$:

```
> x = c(1 ,6 ,2)
> x
[1] 1 6 2
> y = c(1 ,4 ,3)
```

- The *matrix*() function can be used to create a matrix of numbers. Before matrix() we use the matrix() function, we can learn more about it:

```
> ? matrix
> x= matrix (data=c(1 ,2 ,3 ,4),nrow=2,ncol=2)
> x
[,1] [,2]
[1,] 1 3
[2,] 2 4
```

Loading Data

For most analyses, the first step involves importing a data set into R. The read.table() function is one of the primary ways to do this. The help file read.table() contains details about how to use this function. We can use the function write.table() to export data

Once the data has been loaded, the fix() function can be used to view it in a spreadsheet like window. However, the window must be closed before further R commands can be entered

```
> Auto=read.table (" Auto .data")
> fix ( Auto)
```

```
> Auto=read.table("Auto .data",header =T,na.strings ="
> fix ( Auto)
```

Loading Data

Excel is a common-format data storage program. An easy way to load such data into R is to save it as a csv (comma separated value) file and then use the read.csv() function to load it in.

```
> Auto=read.csv("Auto.csv",header =T,na.strings ="?")
> fix ( Auto)
> dim ( Auto)
[1] 397 9
```

The dim() function tells us that the data has 397 observations, or rows, and 9 variables, or columns.